Inheritance

39A

In this exercise you will study the inheritance of four characteristics, or traits, of some mice-like creatures. You will need to revise your knowledge of the following technical terms used in genetics: cross, pure-breeding, genotype, phenotype, offspring, F_1 generation, F_2 generation, homozygous, heterozygous, dominant, recessive, and Punnett squares.

The table below shows the selected traits.

Note: ♂ = male ♀ = female

Characteristic	Dominant		Recessive	
Tail shape	T	straight	t	curly
Ear shape	E	round	e	oval
Coat type	C	smooth	c	rough
Whisker length	W	long	w	short

Using Punnett squares, find the ratios of the genotypes and, hence, the phenotypes of the offspring of the crosses. Draw the appropriate traits (phenotypes) on the 'mouse' pictures.

Cross 1

What are the genotypes and phenotypes of the F_1 generation resulting from a cross between a pure-breeding, straight tailed male and a pure-breeding, curly tailed female?

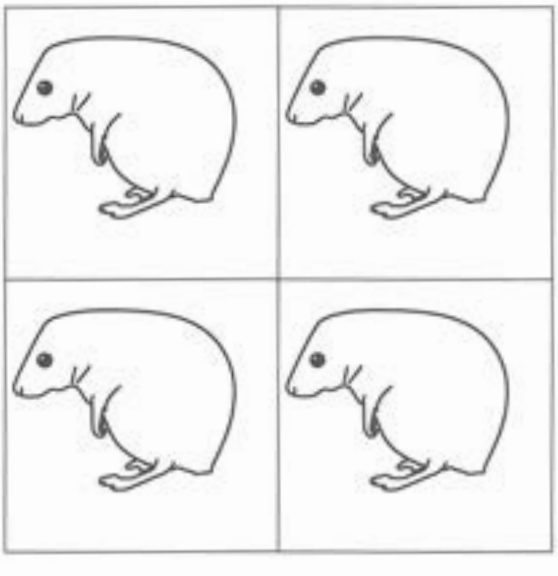
Genotypes (parents): ♂ _____ × ♀ _____



Genotypes of F_1: _____

Phenotypes of F_1:

_____



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Cross 2

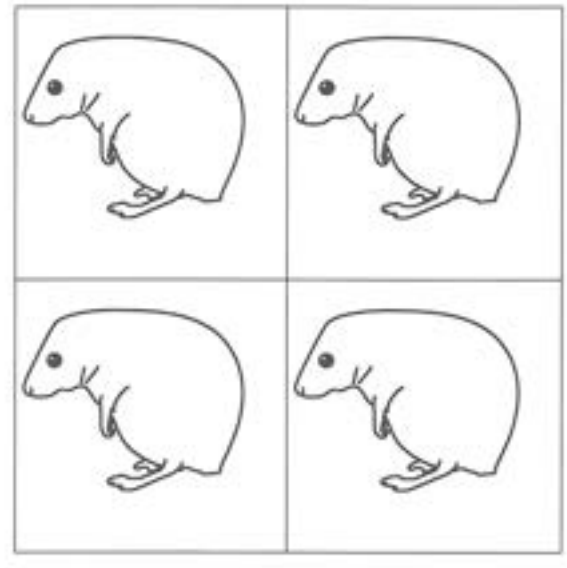
A male and a female from 'cross 1' were allowed to breed. What are the genotypes and phenotypes of the $F_1 \times F_1$ cross?

Genotypes (parents) F_1: ♂ _____ × ♀ _____



Genotypes of F_2: _____

Phenotypes of F_2: _____



Cross 3

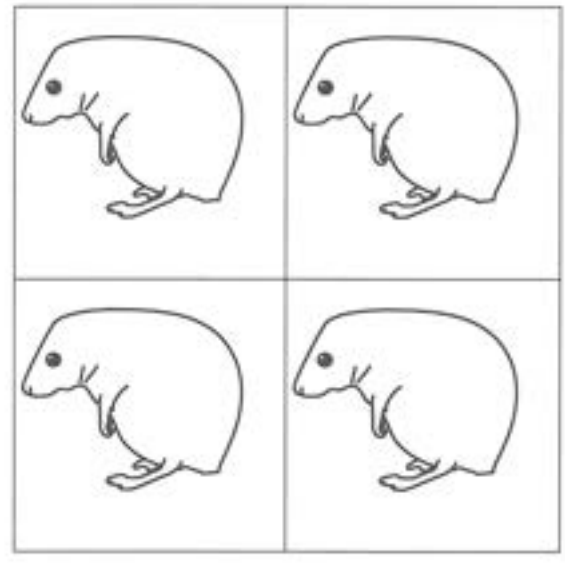
Consider a cross between a homozygous oval-eared male and a homozygous round-eared female. What are the genotypes and phenotypes of this cross?

Genotypes (parents): ♂ _____ × ♀ _____



Genotypes of F_1: _____

Phenotypes of F_1: _____



Cross 4

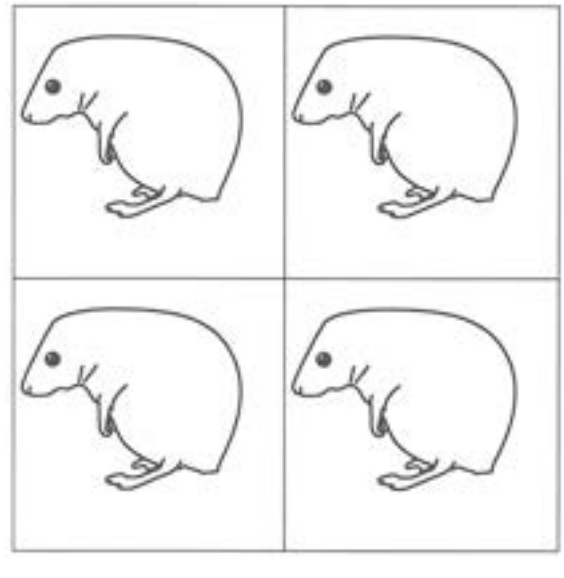
The F_1 offspring from 'cross 3' are allowed to breed. What are the possible genotypes and phenotypes of the F_2 generation?

Genotypes (parents) F_1: ♂ _____ × ♀ _____



Genotypes of F_2: _____

Phenotypes of F_2: _____



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Cross 5

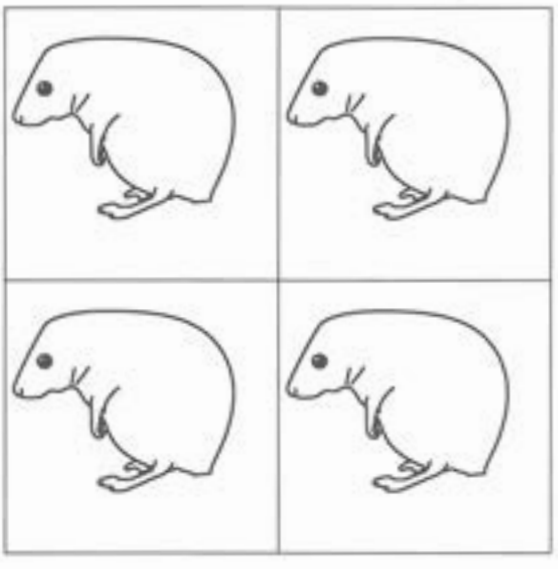
Consider a cross between a heterozygous long-whiskered male and a homozygous short-whiskered female. What are the genotypes and phenotypes of this cross?

Genotypes (parents): ♂ _____ × ♀ _____



Genotypes of F_1: _____

Phenotypes of F_1: _____



Cross 6

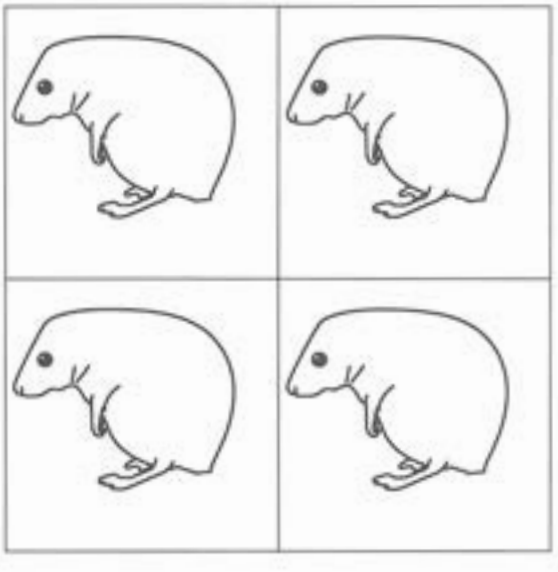
Consider a cross between a homozygous rough-coated male and a homozygous smooth-coated female. What are the genotypes and phenotypes of this cross?

Genotypes (parents): ♂ _____ × ♀ _____



Genotypes of F_1: _____

Phenotypes of F_1: _____



Cross 7

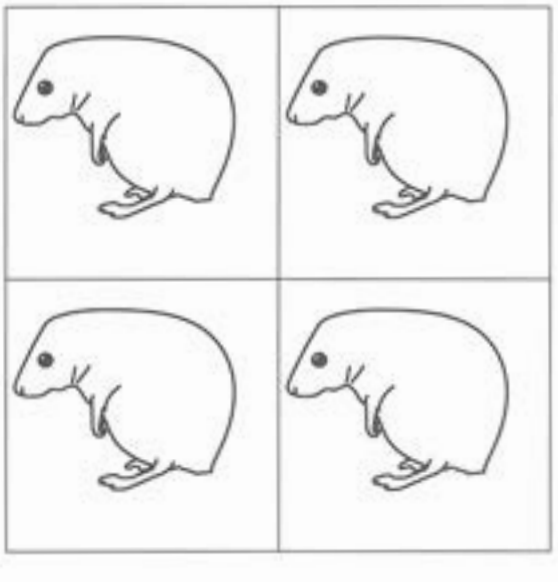
The offspring of 'cross 6' are allowed to interbreed. What are the genotypes and phenotypes of this cross?

Genotypes (parents) F_1: ♂ _____ × ♀ _____



Genotypes of F_2: _____

Phenotypes of F_2: _____



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